Wisdom Bioscience
166 N Waverly Street, Orange, CA 92866 | (555) 555-5555

PATIENT & SPECIMEN INFORMATION

Patient name	Chen, David W.	**Accession**	NGS-2026-052101
Date of birth	1961-07-22 (Age 64)	**Specimen**	Oral swab, left lateral tongue
MRN	10345621	**Collected**	2026-05-19
Ordering physician	Dr. Sarah Chen, DDS MD	**Received**	2026-05-19
Clinical indication	Oral lesion — left lateral tongue	**Reported**	2026-05-20

RESULT: POSITIVE

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
KRAS	NM_004985.5:c.35G>T	p.Gly12Asp	0.43%	463×	I	Pathogenic
PIK3CA	NM_006218.4:c.3140A>G	p.His1047Arg	0.70%	714×	I	Pathogenic

CLINICAL INTERPRETATION

KRAS p.Gly12Asp | Pathogenic (Tier I)

KRAS p.Gly12Asp (G12D) is one of the most frequently detected oncogenic driver mutations in human cancer. It locks KRAS in the GTP-bound active state, constitutively activating downstream RAS/MAPK and PI3K/AKT/mTOR proliferative signalling. In oral squamous cell carcinoma, KRAS mutations are found in 3–8% of cases and mark a subgroup with aggressive behaviour. This variant is classified Pathogenic (Tier I, COSMIC ID COSM521). The co-occurrence with PIK3CA H1047R (see below) indicates dual activation of RAS and PI3K pathways, a combination associated with resistance to single-agent targeted therapies and may influence enrolment eligibility in clinical trials. NOTE: Observed VAF (0.43%) is near the assay reporting threshold; this call is supported by 2 high-quality consensus reads and has been reviewed by the laboratory director. Orthogonal confirmation is recommended before clinical action.

PIK3CA p.His1047Arg | Pathogenic (Tier I)

PIK3CA p.His1047Arg is a hotspot activating mutation in the kinase domain of the p110α catalytic subunit of PI3-kinase. It constitutively activates the PI3K/AKT/mTOR pathway, promoting cell survival, proliferation, and tumour angiogenesis. PIK3CA mutations occur in 15–20% of OSCC, most commonly at E545K and H1047R hotspots. This variant is classified Pathogenic (Tier I, COSMIC ID COSM775). PIK3CA H1047R is potentially actionable: alpelisib (PI3Kα inhibitor) is FDA-approved for PIK3CA-mutated breast cancer and is under investigation in HNSCC. Multidisciplinary tumour board review for consideration of PI3K-directed therapy or clinical trial enrolment is recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, NFE2L2, NOTCH1, NSD1, PIK3R1, PTEN, TERT, TP53
Variants of uncertain significance (Tier III) are not reported.

METHOD SUMMARY

Cell-free and cellular DNA extracted from oral swab is subjected to hybridization capture using the IDT xGen Duplex UMI panel targeting 251 genomic regions (~77 kbp) across 21 cancer-relevant genes. Libraries incorporate 9-bp Unique Molecular Identifiers (UMIs) in a 9M1S+T read structure on both R1 and R2. Libraries are sequenced on Illumina NextSeq (2×150 bp). After alignment to GRCh38 (canonical chromosomes), reads are grouped by UMI using fgbio (adjacency strategy, ≤1 edit distance), and molecular consensus sequences are called requiring ≥2 reads per UMI family. Somatic variants are called on the consensus BAM with VarDict at minimum allele frequency 0.001. Variants ≥0.5% consensus VAF with ≥2 consensus supporting reads are reported. UMI deduplication reduces sequencing noise by >800-fold. Run-level negative controls are included on every sequencing run.

LIMITATIONS

This assay detects somatic single nucleotide variants and small indels within 251 target regions (~77 kbp) at VAF ≥0.5% with validated sensitivity of 98.4% at ≥20,000× raw input depth (≥400× consensus depth after UMI deduplication). Structural variants, copy number alterations, gene fusions, large indels, intronic variants outside defined capture regions, and variants in regions with low hybridization capture efficiency are not reliably assessed. A negative result does not exclude malignancy. Tumour heterogeneity, low tumour cell content, and specimen quality may affect variant allele fraction. Variants below the 0.5% VAF reporting threshold may be present but are not reported. At 2,000× raw sequencing depth, approximately 15–20% of targets may fall below 100× consensus depth; a result of no variant detected at low-coverage targets should be interpreted with caution. Results should be interpreted in the context of clinical and pathological findings by a qualified clinician.

Reported by: Dr. A. Patel, MD PhD, FCAP **Report date: 2026-05-20**

Results reviewed and approved by the Laboratory Director. This report is electronically signed and does not require a wet signature.

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PATIENT & SPECIMEN INFORMATION

Patient name	Williams, Margaret A.	**Accession**	NGS-2026-052102
Date of birth	1970-03-11 (Age 56)	**Specimen**	Oral swab, right buccal mucosa
MRN	20456732	**Collected**	2026-05-19
Ordering physician	Dr. Marcus Reid, MD	**Received**	2026-05-19
Clinical indication	Oral lesion — right buccal mucosa	**Reported**	2026-05-20

RESULT: POSITIVE

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
NOTCH1	NM_017617.5:c.7244C>T	p.Pro2415Leu	0.26%	769×	II	Likely Pathogenic

CLINICAL INTERPRETATION

NOTCH1 p.Pro2415Leu | Likely Pathogenic (Tier II)

NOTCH1 functions as a tumour suppressor in stratified squamous epithelium. The p.Pro2415Leu missense variant affects a conserved residue in the PEST domain, predicted to disrupt protein stability (SIFT: deleterious; PolyPhen-2: probably damaging). NOTCH1 loss-of-function mutations are found in 10–15% of OSCC and are associated with early-onset tumourigenesis in the oral cavity. This variant is classified Likely Pathogenic (Tier II). In the context of confirmed dysplasia, detection of a NOTCH1 tumour suppressor variant supports malignant progression risk. Clinical correlation and multidisciplinary review are recommended. NOTE: Observed VAF (0.26%) is below the assay nominal 0.5% reporting threshold. This call is at the limit of detection (VD = 2 consensus reads at 769× consensus depth) and has been reviewed and approved by the laboratory director. Repeat testing at higher sequencing depth or orthogonal confirmation is strongly recommended before clinical action.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NSD1, PIK3CA, PIK3R1, PTEN, TERT, TP53
Variants of uncertain significance (Tier III) are not reported.

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Reported by: Dr. A. Patel, MD PhD, FCAP **Report date: 2026-05-20**

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PATIENT & SPECIMEN INFORMATION

Patient name	Thompson, Richard E.	**Accession**	NGS-2026-052103
Date of birth	1951-09-28 (Age 74)	**Specimen**	Oral swab, left floor of mouth
MRN	30567843	**Collected**	2026-05-19
Ordering physician	Dr. Sarah Chen, DDS MD	**Received**	2026-05-19
Clinical indication	Oral lesion — left floor of mouth	**Reported**	2026-05-20

RESULT: POSITIVE

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.60%	664×	I	Pathogenic

CLINICAL INTERPRETATION

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His is the most frequently observed p53 mutation in human cancer. It disrupts the structural zinc-binding domain of the DNA-binding domain, abolishing wild-type tumour suppressor function, while also conferring dominant-negative and gain-of-function oncogenic properties including promotion of invasion, metastasis, and MDM2 antagonism (COSMIC ID COSM10648). In OSCC, TP53 mutations are present in 50–70% of cases and are associated with aggressive tumour behaviour, locoregional recurrence, and reduced sensitivity to platinum-based chemotherapy. The VAF of 0.60% in a surveillance swab warrants clinical correlation to exclude early local recurrence. Imaging and clinical re-evaluation are recommended.

GENES WITH NO PATHOGENIC OR LIKELY PATHOGENIC VARIANTS DETECTED

AJUBA, CASP8, CDKN2A, CUL3, DPYD, EGFR, EP300, FAT1, FBXW7, HRAS, KEAP1, KMT2D, KRAS, NFE2L2, NOTCH1, NSD1, PIK3CA, PIK3R1, PTEN, TERT

Variants of uncertain significance (Tier III) are not reported.

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Reported by: Dr. A. Patel, MD PhD, FCAP **Report date: 2026-05-20**

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PATIENT & SPECIMEN INFORMATION

Patient name	Garcia, Ana L.	**Accession**	NGS-2026-052104
Date of birth	1967-12-04 (Age 58)	**Specimen**	Oral swab, soft palate
MRN	40678954	**Collected**	2026-05-19
Ordering physician	Dr. Linda Park, MD	**Received**	2026-05-19
Clinical indication	Oral lesion — soft palate	**Reported**	2026-05-20

RESULT: POSITIVE

DETECTED VARIANTS

Gene	HGVS (coding)	Protein change	VAF	Depth	Tier	Classification
EGFR	NM_005228.5:c.2573T>G	p.Leu858Arg	0.57%	525×	I	Pathogenic
TP53	NM_000546.6:c.524G>A	p.Arg175His	0.30%	669×	I	Pathogenic

CLINICAL INTERPRETATION

EGFR p.Leu858Arg | Pathogenic (Tier I)

EGFR p.Leu858Arg (L858R) is a well-characterised activating mutation in exon 21 of the EGFR kinase domain. It disrupts the autoinhibitory conformation, leading to constitutive EGFR kinase activation and downstream RAS/MAPK and PI3K/AKT signalling. While canonical in non-small cell lung adenocarcinoma (where it predicts response to EGFR TKIs such as osimertinib), EGFR L858R has been reported in 1–3% of OSCC. Cetuximab (anti-EGFR monoclonal antibody) is FDA-approved in HNSCC and may have enhanced activity in the setting of EGFR activating mutations. Multidisciplinary tumour board review for consideration of EGFR-directed therapy or clinical trial enrolment is recommended. Co-occurring TP53 R175H (see below) may affect response to EGFR-directed therapy and warrants combined consideration.

TP53 p.Arg175His | Pathogenic (Tier I)

TP53 p.Arg175His — see interpretation above (Thompson, Richard E., NGS-2026-052103). Co-occurrence of EGFR L858R and TP53 R175H has been associated with resistance to EGFR-directed therapies in other tumour types; multidisciplinary tumour board review is recommended. NOTE: Observed VAF (0.30%) is below the assay nominal 0.5% reporting threshold. This call is at the limit of detection (VD = 2 consensus reads at 669× consensus depth) and has been reviewed and approved by the laboratory director. Orthogonal confirmation is recommended before clinical action.

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